

PHYSICAL LAYER STANDARDS SURVEY & SIMULATION

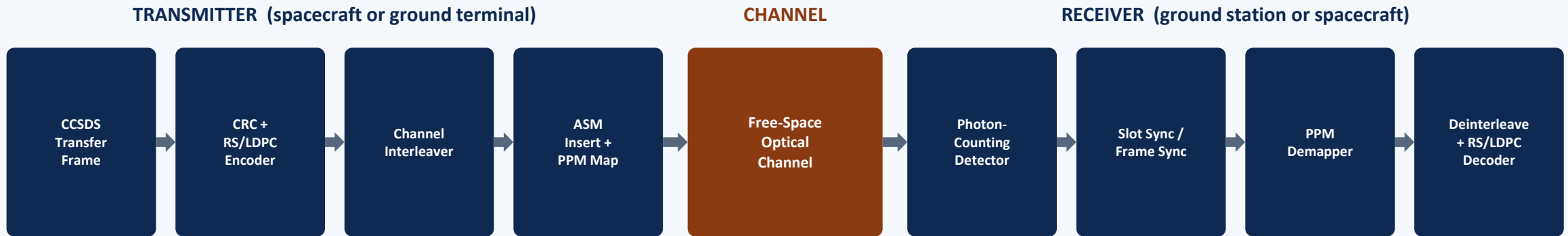
CCSDS Optical Space Communication

CCSDS 141.0-B-1 Optical Communications Physical Layer | **CCSDS 142.0-B-1** Optical Communications Coding & Synchronization

Group: [Your Name(s)] **Track:** Space / GIS **Date:** 25 August 2026

2026 Standards Survey and Simulation (Physical Layer) — Presentation Deck

Overall Block Diagram: Transmitter → Channel → Receiver



CCSDS 142.0-B-1 (Coding & Sync Sublayer)

CCSDS 141.0-B-1 (Physical Layer — laser, slots, OOK)

How the two terminals interact

- Ground Terminal A continuously transmits an uplink beacon (+ optional AOS/USLP command frames).
- Spacecraft Terminal B locks its Fine Steering Mirror onto the beacon for pointing, acquisition & tracking, then transmits downlink telemetry.
- Both directions pass through the same coding chain: CRC → RS(255,223) or LDPC (rate 1/2, 9/10) → interleaver → PPM slot mapping.

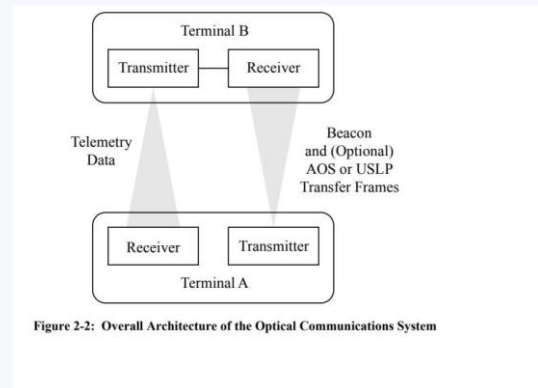
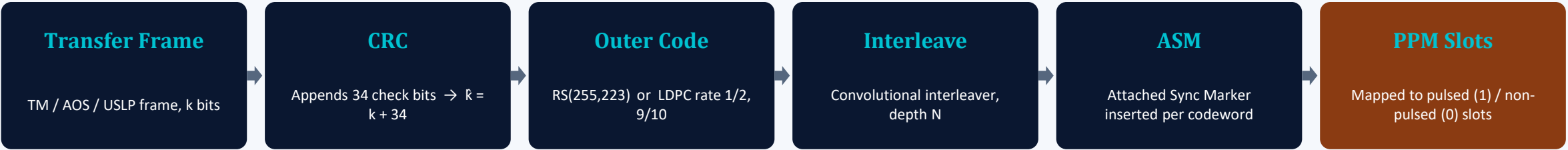


Fig. 2-2, CCSDS 141.0-B-1 — Terminal A / Terminal B link roles

Framing: From Transfer Frame to Pulsed Slots

The Coding & Sync sublayer (142.0-B-1) wraps each CCSDS Transfer Frame before the Physical Layer (141.0-B-1) ever sees a bit:



Managed Physical-Layer Parameters — Table 5-1, CCSDS 141.0-B-1

Parameter	Allowed Values
Telemetry center-frequency index n	−18 to 28 → 193.1 + n×0.1 THz (C-band, 100 GHz grid)
Telemetry signaling slot width	0.125, 0.25, 0.5, 1, 2, 4, 8, 16, 32, 64, 128, 256, or 512 ns
Telemetry polarized transmission	Used / Not Used (RHCP, extinction ratio > 10 dB)
Beacon frequency	280.18, 281.72, or 291.06 THz (1070 / 1064.15 / 1030 nm)
Data transmission on beacon	Used / Not Used

Modulation: Non-Binary PPM at Coding Layer → Binary OOK at Physical Layer

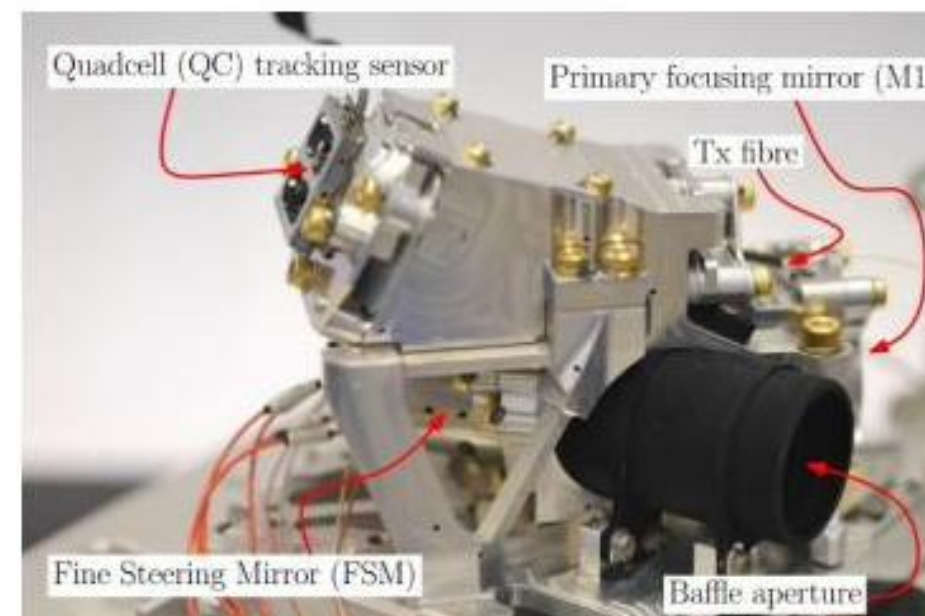
M-ary PPM *(non-binary, at 142.0-B-1)*

Each symbol occupies one of M time slots; the laser pulses in exactly one slot per symbol. Chosen for photon efficiency: deep-space / photon-starved links need the fewest photons per bit, not the highest raw baud rate.

OOK-NRZ *(binary, at the Physical Layer / 141.0-B-1)*

Each PPM slot is realized physically as a binary intensity level: laser ON (1) or OFF (0). High-data-rate near-Earth links (e.g. CubeCAT, 1 Gbps) run OOK-NRZ directly, without the PPM layer — trading photon efficiency for simplicity and raw throughput.

advanced dampers [4] to enable the CubeCAT terminal to comply with the missions launch requirements [5].



CubeCAT 1U optical terminal — Fine Steering Mirror & tracking sensor for PAT, transmit fibre feeding the OOK-modulated laser (Broekens et al., ICSOS 2025)

1545 nm downlink (OOK) · 1590 nm uplink beacon · 1 Gb/s PRBS23 data rate

Channel: Shot Noise, Atmosphere & Pointing Loss

Shot / Photon-Counting Noise

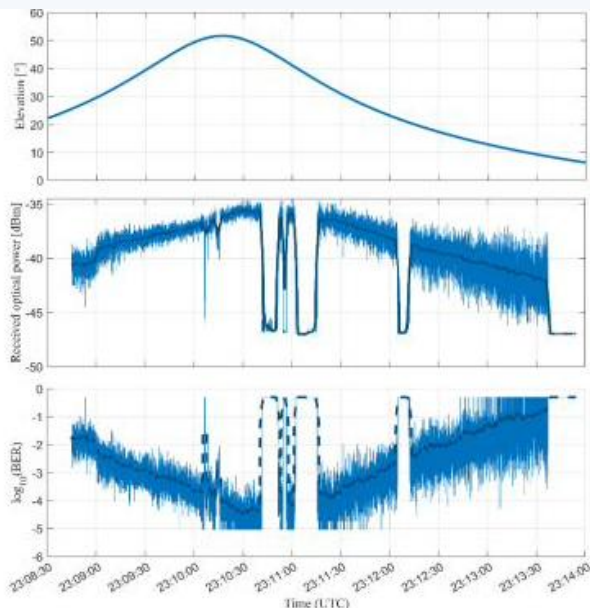
Free-space optical receivers are photon-limited: detection statistics follow a Poisson process rather than classic AWGN, especially in the HPE (photon-starved) regime targeted by 141.0-B-1.

Atmospheric Turbulence

Scintillation and beam wander from the atmosphere cause fast fading of received power — visible directly as the noisy envelope riding on the slow elevation-angle trend below.

Geometric / Pointing Loss

Received power scales with slant range and elevation angle; a PAT subsystem (beacon-tracking Fine Steering Mirror) keeps the beam aligned as geometry changes across a pass.



Elevation, received optical power, and $\log_{10}(\text{BER})$ across a full LEO overpass — MeO Optical Ground Station, 6 Dec 2024 (Broekens et al., ICSOS 2025).

Notice how BER tracks received power, which itself tracks elevation-driven path loss and turbulence: the two brief power dropouts near 23:10–23:12 UTC directly produce visible BER spikes.

Demodulation: Photon-Counting Detection & Slot Sync

1

Optical Front-End

Telescope + APD / photon-counting detector converts incident photons to an electrical current proportional to slot energy.

2

Slot & Frame Sync

Receiver locks to the Attached Sync Marker to recover slot timing and codeword boundaries before any decoding can start.

3

PPM / OOK Demapper

For OOK: threshold each slot to a 0/1 decision (or pass soft values). For PPM: pick the slot(s) with highest energy per M-slot symbol.

4

Soft-Decision Output

Slot energies (not just hard bits) are handed to the RS/LDPC decoder — soft information meaningfully improves coding gain.

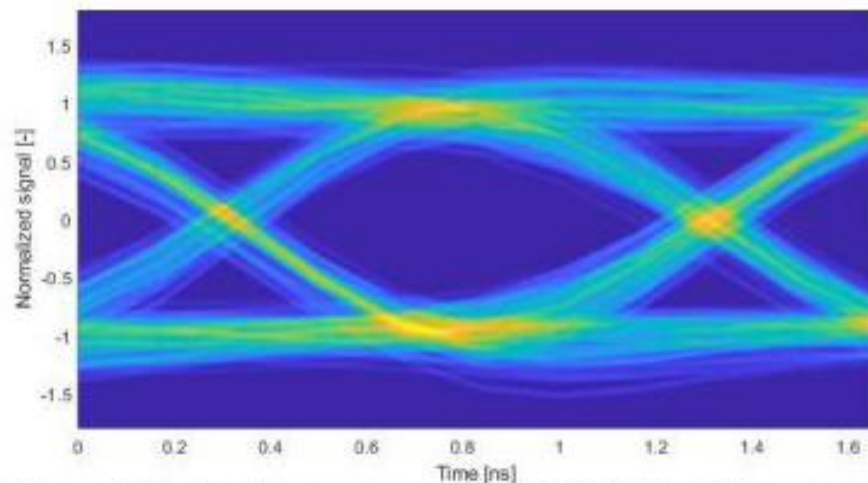


Figure III-1: Eye diagram containing 1000 digitized data traces measured by the MeO detector near culmination

Reading the eye diagram

1000 overlaid data traces from the MeO Optical Ground Station receiver, captured near the pass culmination point (highest elevation, strongest signal).

A wide, clearly-open eye confirms the OOK threshold decision (step 3 above) has enough margin to separate a '10' from a '0', even after atmospheric turbulence and detector bandwidth-limiting (600 MHz APD) round the pulse shape into the sinusoidal form seen here.

ECC: Reed–Solomon / LDPC Outer Code + SCPPM Inner Code

SCPPM Encoder Structure (142.0-B-1 §3)

Input block $R = 15120r$ bits (k data bits + 34-bit CRC)

Outer code Rate-1/3 convolutional “mother code”

Interleave Bit interleaver decorrelates burst errors

Accumulate Differential accumulator (rate-1 code)

PPM mapping 15120 coded bits $\rightarrow$ M-ary PPM symbols

Initialized to the all-zeros state before each input block (self-terminating via CRC/termination digits).

Outer Block Code Options

Reed–Solomon (255, 223)

8-bit symbols · corrects up to 16 symbol errors per codeword

LDPC, rate 1/2

higher coding gain, more decoder complexity

LDPC, rate 9/10

low overhead for high-SNR / high-photon-flux links

Decoder (Receiver Side — mirrors the encoder)

- De-interleave soft PPM slot-energy metrics
- Iterative / Viterbi decode of inner convolutional-accumulate code
- RS or LDPC outer decode corrects residual symbol errors
- CRC check confirms frame integrity before hand-off to Data Link layer

Real-World Validation: CubeCAT / SmallCAT In-Orbit Link

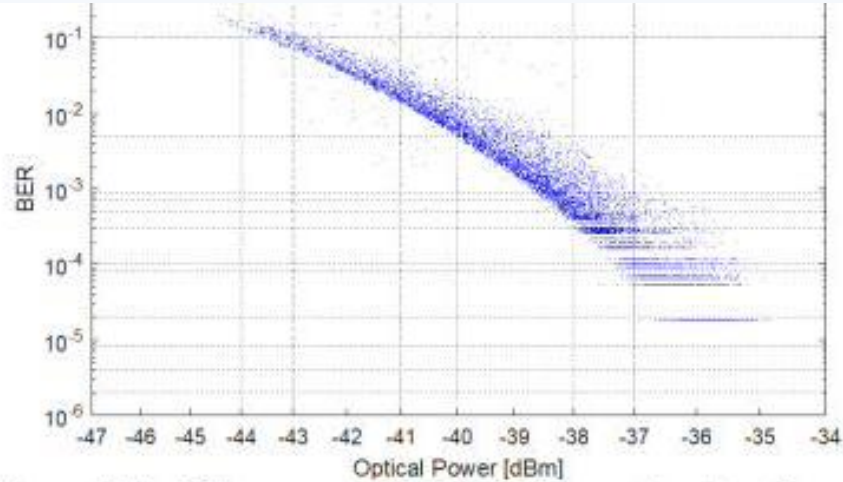


Figure III-3: BER versus received optical power based on the digitized MeO data signal

Fig. III-3 — BER vs. received optical power, MeO OGS, offline-processed downlink (Broekens et al., ICSOS 2025)

1 Gbps DTE optical downlink — two independent ground stations

-44 → -35 dBm

MeO received power range

$2 \times 10^{-2} \rightarrow 1 \times 10^{-6}$

MeO waterfall BER range

155 s

TNO OGS link duration

$< 1 \times 10^{-6} \rightarrow$ error-free

TNO OGS BER after acquisition

Takeaway: BER improves as a clean waterfall with received optical power, exactly as predicted by the photon-counting channel model — confirming CCSDS-compliant OOK/PAT design closes real LEO-to-ground links, even before FEC is applied.

Summary & References

- CCSDS 141.0-B-1 defines the optical Physical Layer (laser frequency, linewidth, OOK modulation, slot width) for HPE telemetry and beacon links.
- CCSDS 142.0-B-1 defines the Coding & Synchronization sublayer directly above it: CRC, RS(255,223)/LDPC outer coding, SCPPM inner coding, and PPM slot mapping.
- Photon-counting statistics dominate the channel model — atmospheric turbulence and pointing/elevation geometry drive slow power fading on top of that.
- In-orbit CubeCAT/SmallCAT results show CCSDS-aligned OOK links closing a real LEO pass with clean BER waterfalls down to error-free operation.

Thank you — Q&A

References

- [1] CCSDS 141.0-B-1, Optical Communications Physical Layer, Recommended Standard, Issue 1. Washington, D.C.: CCSDS, Aug. 2019.
- [2] CCSDS 142.0-B-1, Optical Communications Coding and Synchronization, Recommended Standard, Issue 1. Washington, D.C.: CCSDS, Aug. 2019.
- [3] K. Broekens et al., “In-orbit Test Results of a 1 Gbps Free Space Direct-to-Earth Optical Satellite Link of a CubeSat-compatible Laser Communication Terminal,” 2025 IEEE Int. Conf. on Space Optical Systems and Applications (ICSOS), 2025.